



Influence of solvent chemistry and pre-annealing temperature on the structural optical and electrical properties of solution-processed CTS thin films

Sevde Erkan^{1,2}, Yavuz Atasoy^{1,3}, M. Ali Olgar^{1,2}, and Recep Zan^{1,2,*}

¹ Nanotechnology Application and Research Center, Niğde Ömer Halisdemir University, Niğde 51200, Turkey

² Department of Physics, Niğde Ömer Halisdemir University, Niğde 51200, Turkey

³ Niğde Zübeyde Hanım Vocational School of Health Services, Niğde Ömer Halisdemir University, Niğde 51200, Turkey

Received: 24 November 2025

Accepted: 1 April 2026

Published online:

9 April 2026

© The Author(s), 2026

ABSTRACT

In this study, Cu₂SnS₃ (CTS) thin films were synthesized using two different solution types: an organic solvent 2-methoxyethanol (2ME) and 2-methoxyethanol/distilled water (2-ME/DW), and the effects of solution type and pre-annealing temperature (200, 250, and 300 °C) on the structural and electrical properties of CTS films were systematically investigated. The solvent type influences precursor decomposition and crystallization behavior. Structural analysis revealed that the sample prepared with 2-ME (Type-I CTS) and pre-annealed at 200 °C (CTS-I-200) exhibited the largest crystallite size (22.52 nm) and the lowest values of microstrain (1.6×10^{-3}), dislocation density ($1.97 \times 10^{-3} \text{ nm}^{-2}$), and grain boundary density (1.14 nm^{-2}), which are beneficial for photovoltaic performance due to improved charge transport and reduced recombination losses. Raman and XRD results confirmed formation of monoclinic CTS phase and indicated that the CuS phase tends to form in the bulk region rather than on the surface. Additionally, pre-annealing at 200 °C led to a more homogeneous grain distribution, with large grains integrating into the film structure. Elemental analysis showed that all samples had Cu-rich composition regardless of solution type and pre-annealing treatment. With increasing temperature, the sulfur content decreased in Type-I CTS samples but increased in aqueous-based (Type-II CTS) samples, indicating more effective sulfur retention in the presence of distilled water. The optical bandgap (E_g) varied slightly in Type-I CTS films (0.98–1.02 eV) and more significantly in Type-II CTS films (0.99–1.07 eV), influenced by Cu/Sn ratio and secondary phases like SnS₂. All films showed p-type conductivity. The CTS-I-200 film showed $128 \times 10^{17} \text{ cm}^{-3}$ carrier concentration and 3.96 $\Omega \cdot \text{cm}$ resistivity values, while Type-II CTS films exhibited higher carrier concentration and lower resistivity due to the presence of Cu-rich secondary phases. Overall, it was concluded that both the solution type and pre-annealing temperature have a significant impact on the structural, optical and electrical properties of CTS thin films. Based

Address correspondence to E-mail: recep.zan@ohu.edu.tr

on all the analyses performed, the CTS-I-200 sample, produced using the 2-ME solution and pre-annealed at 200 °C, was found to exhibit superior structural properties for potential solar cell applications.

1 Introduction

Photovoltaic (PV) technology is a field of intense research today, offering an environmentally friendly energy solution. While silicon-based solar cells represent a mature technology and dominate the current PV market, alternative thin-film technologies like Copper Indium Gallium Selenide (CIGS) and Cadmium Telluride (CdTe) are emerging as compelling sustainable energy solutions, driven by their economic advantages and wide-ranging potential applications [1]. However, these technologies also face some challenges such as the scarcity of indium and gallium, and the toxicity of cadmium (Cd) [2]. For these reasons, researchers have focused on developing alternative solar cell compounds made of abundant and environmentally friendly elements. In this context, copper-based quaternary/pentenary compounds such as $\text{Cu}_2\text{ZnSn}(\text{S},\text{Se})_4$ (CZTSSe), which crystallize in the kesterite phase, show promising performance potential due to their optimal optical bandgap, p-type conductivity, and high absorption coefficient ($>10^4 \text{ cm}^{-1}$) [3, 4]. Although the conversion efficiency of CZTS(e)-based solar cells has already reached 15.1%, challenges arise in obtaining a single kesterite phase-free from secondary phases due to the large number of elements it contains [5]. Considering the difficulties in synthesizing the pure CZTSSe kesterite phase, the ternary Cu-based Cu_2SnS_3 (CTS) chalcogenide compound, which has a simpler structure, appears to be an attractive alternative in this regard [6]. The relative simplicity of CTS, combined with a significant reduction in undesirable secondary phases such as zinc sulfide (ZnS) and copper/zinc (Cu/Zn) disorder, establishes it as a promising candidate for thin-film photovoltaic applications [7]. Furthermore, CTS exhibits p-type semiconductor behavior, high mobility, and a high optical absorption coefficient ($>10^4 \text{ cm}^{-1}$), making this semiconductor material particularly suitable for use in solar cells and other optoelectronic technologies [8]. Furthermore, the bandgap of CTS is directly correlated with its crystal structure

(monoclinic, cubic, tetragonal) and varies from 0.9 to 1.7 eV. Among the crystal phases of CTS thin films, the monoclinic phase demonstrates superior solar cell performance due to its more ordered cation arrangement, which results in fewer antisite defects and vacancies compared to the cubic or tetragonal phases [9].

Various physical and chemical techniques are used in the fabrication of CTS thin films comprising chemical bath deposition [10], sputtering [11], spray pyrolysis [12], successive ionic layer adsorption and reaction (SILAR) [13] thermal evaporation, pulsed laser deposition (PLD) [14], electrodeposition [15, 16], and spin-coating [17]. Thin films produced through physical methods offer advantages like high purity, homogeneity, and easy thickness control. However, the requirement for a vacuum environment results in higher time and energy consumption, ultimately leading to increased production costs. On the other hand, chemical methods offer simple, low-cost, and rapid production opportunities for CTS thin films. Among these methods, the spin-coating method is one of the very commonly preferred methods due to both the advantage of the solution-based process and the record efficiency (~ 15%) obtained in CZTS by this method, indicating the usability of this method in CTS growth [5].

For the reasons mentioned above, CTS thin films were fabricated using the spin-coating method in the present study. According to the reported literature, CTS-based solar cells fabricated by spin-coating have achieved a conversion efficiency of around 3% [18]. However, the existing literature highlights that the efficiency of CTS-based solar cells is still inadequate, considering that the Shockley–Queisser limit predicts a conversion efficiency of around 30% for CTS-based p–n junctions. To address this gap, significant research efforts have been directed toward fabricating CTS thin films with enhanced crystalline quality and morphology. The properties of these thin films are critically determined by the growth parameters including sulfurization temperature, sulfurization

time, pre-annealing, chemical sources and solvent type [19]. Among these factors, solvents play a critical role in determining the quality of CTS thin films. Various organic (such as alcohols and their derivatives) and inorganic (such as distilled water) solvents have been employed in the fabrication of CTS thin films. Some reported studies on CTS thin films produced using different solvents along with various sulfurization parameters and their effects on the properties of the resultant films are provided in Table 1. As evident from this table, a large number of attempts have been carried out in order to achieve the optimal CTS film. For instance, Dias et al. demonstrated that CTS thin films produced by dissolving the chemicals in 2-methoxyethanol (2-ME) exhibited superior structural and optical properties, highlighting the potential use of CTS in solar cells and other optoelectronic devices [20]. On the other hand, Sayed et al. investigated the effects of a water-based solvent on the structural, morphological, and photovoltaic properties of CTS thin films produced using the chemical spray pyrolysis method [21]. They showed that the CTS absorber layers had a morphology with a void-free, fine-grained structure, which enhanced the device's performance. Research highlights the impact of solvent selection on CTS thin-film properties. Therefore, exploring solvent mixtures to achieve a synergistic effect for an optimal CTS absorber layer is a promising approach. However, there are only a limited number of studies that discuss the effects of solvent mixtures in varying proportions (see Table 1). In a study conducted by Suryawanshi et al., CTS thin films were obtained by a spin-coating system using a mixture of distilled water (DW)/ethanol (5:1) in solution [22]. It has been evaluated that the aqueous solution improves the grain structure and crystalline quality. The presence of DW in the solvent facilitates the formation of a void-free microstructure, indicating improved morphology. In another study conducted by Chen et al., nanostructured CTS was synthesized using the solvothermal method with varying ethanol concentrations [23]. The study demonstrated that modifying the ethanol content led to the formation of CTS nanostructures exhibiting both cubic and tetragonal crystal structures. Furthermore, the bandgap energy of the tetragonal CTS nanocrystals synthesized in a 50% ethanol–water solution was found to be close to the optimal value for photovoltaic solar energy conversion in single junction devices. Overall, the limited research on solvent mixtures indicates a synergistic effect in aqueous-based systems.

Another effective method for CTS thin films and their use in solar cell optimization is the pre-annealing approach. The pre-annealing treatment plays a key role among the sulfurization parameters. Chalapathy et al. developed a process using a two-step pre-annealing temperature to produce a void-free CTS film with a smooth surface and tightly bonded grains [24]. In another study, by using the spin-coating method and repeating the pre-annealing process several times to obtain the appropriate thickness, monoclinic CTS thin films with compact bulk and large grain size were reported to be obtained at 540 °C sulfurization temperature [25]. Research conducted in this direction suggests that pre-annealing is an effective strategy for enhancing the structural and optoelectronic properties of CTS thin films, thereby improving solar cell performance.

Although CTS thin films have been widely investigated, most existing studies address individual processing parameters in isolation, such as solvent choice or thermal treatment, rather than examining their combined influence. In particular, there is a lack of systematic studies that directly compare pure organic and aqueous–organic solvent systems together with controlled pre-annealing treatments. The effects of moderate pre-annealing temperatures applied prior to high-temperature sulfurization, especially in spin-coated, solution-processed CTS films, remain insufficiently understood. Moreover, how precursor solution chemistry and pre-annealing interact with rapid thermal sulfurization to influence phase formation, grain growth, and film morphology has not yet been clearly established. Clarifying these relationships is crucial for developing reliable processing strategies and for improving the quality and performance of CTS absorber layers in thin-film photovoltaic devices.

In the present study, the authors comparatively investigated the effects of solution type—using either pure 2-mercaptoethanol (2-ME) or an aqueous mixture (DW/2-ME)—along with pre-annealing treatments at 200, 250, and 300 °C, on the properties of solution-processed CTS thin films. A two-step method was employed for film fabrication: spin-coating of precursor layers followed by sulfurization in a rapid thermal processing (RTP) system. The RTP system enabled rapid attainment of high sulfurization temperatures, thereby minimizing the risk of decomposition reactions. This study specifically explores how pre-annealing treatments within the 200–300 °C range, applied prior to high-temperature sulfurization, influence the

Table 1 A review of selected studies from the past decade on chemically deposited CTS films prepared using various solvents

Solvents	Chemicals (Precursor-Stabilizator)	Method	Sulfurization/Deposition process	Structural and optical features	Year	References
DW	$\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$, $(\text{CH}_3\text{SO}_3)_2\text{Sn}$, NH_2CSNH_2	Spray pyrolysis	-Pre-annealing for 250 °C (120 s) -Sulfurized at 550 °C	- Monoclinic—Tetragonal - Crack-free smooth surface - Eg = not given	2019	[21]
EG	$\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, $\text{CN}_2\text{H}_4\text{S}$, $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$, S	Spin Coating	-Pre-annealing for 350 °C (1 min) -Sulfurized at 550 °C (30 min)	- Monoclinic - Large grain and controllable composition - Eg = 1.06 eV	2021	[25]
DMSO	$\text{Cu}(\text{C}_2\text{H}_3\text{O}_2)_2\text{H}_2$, $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, $\text{CH}_4\text{N}_2\text{S}$	Dip Coating	Pre-annealing for 200 °C (5 min) -Sulfurized at 300–475 °C	- Monoclinic - Smooth and homogeneous surface morphology - Eg = 1.43 eV–1.75 eV	2025	[26]
DW	CuCl , SnCl_4 and $\text{Na}_2\text{S}_2\text{O}_3$	CBD	-Sulfurization temperatures of 200 °C–400 °C	- Tetragonal - Densely packed grains with sub-micron dimensions - Eg = 1.71–1.66 eV	2024	[27]
DW-HCl	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ and $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$	CBD	-Pre-annealing 60 °C (2 h) - Sulfurization at 300 °C (1 h)	- Triclinic - Spherical particles with well-defined boundaries - Eg: 1.35 eV and 1.1.31 eV	2017	[10]
DMSO	$\text{Cu}(\text{OAc})_2\text{H}_2\text{O}$, SnCl_2 , $\text{CH}_4\text{N}_2\text{S}$	Spin Coating	-Pre-annealing for 350 °C (3 min) -Sulfurized for 450–570 °C	- Monoclinic - Large grain structures - Eg = 0.88 eV	2018	[28]
DMF	$\text{Cu}(\text{CH}_3\text{COO})_2 \cdot \text{H}_2\text{O}$, $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, $\text{CH}_4\text{N}_2\text{S}$	Spin Coating	- Pre-annealing 300 °C (5 min) -Sulfurized for 470–560 °C	- Monoclinic with a small amount of Cubic phase - Dense structures with columnar grains - Eg = 1.19 eV and 1.43 eV	2020	[29]
2-ME	CuCl , $\text{SnCl}_4 \cdot 5\text{H}_2\text{O}$, $\text{CH}_4\text{N}_2\text{S}$	Spin Coating	- Pre-annealing for 300 °C (5 min) -Sulfurized for 575 °C (20 min)	- Monoclinic - Uniform and dense large grains - Eg = 0.91 eV and 0.97 eV-	2022	[30]
DMSO	$\text{Cu}(\text{OAc})_2 \cdot 2\text{H}_2\text{O}$, SnCl_2 , $\text{CH}_4\text{N}_2\text{S}$	Spin Coating	- Pre-annealing for 350 °C (2 min) -Sulfurized at 580 °C	- Monoclinic - Large grain - Eg: 1.01 eV	2019	[31]
DMSO- MEA	CuCl_2 , SnCl_2 , $\text{CH}_4\text{N}_2\text{S}$	Spin Coating	- Pre-annealing for 220 °C -Post-annealed at 300 °C (30 min)	- Tetragonal - Nano-scale grains - Eg = 1.7 eV	2023	[17]
DW-HCl	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$	CBD	-Pre-annealing 80 °C (2 h) -Annealed at 400 °C (4 h)	- Monoclinic - Eg = 2.7 to 1.8 eV	2016	[22]
2-ME- MEA	$\text{Cu}(\text{CO}_2\text{CH}_3)_2 \cdot \text{H}_2\text{O}$, $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, S	Spin Coating	-Pre-annealing for 180 °C (5 min) -Sulfurized at 300–500 °C (2 h)	- Tetragonal - Hexagonal or rock-like morphology - Eg = 1.15 eV and 2.1 eV,	2022	[32]

Table 1 continued

Solvents	Chemicals (Precursor-Stabilizator)	Method	Sulfurization/Deposition process	Structural and optical features	Year	References
2-ME- DW	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$, $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ and H_2NCSNH_2	Spin Coating	-Pre-annealing for 250 °C (10 min) -Sulfurized at 300–525 °C(60–180–300 s)	- Monoclinic - Granular, void-free, dense and homogeneous - $E_g = 1.01\text{--}1.10$ eV	2025	[33]
2-ME- DW	$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$, $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, H_2NCSNH_2	Spin Coating	-Pre-annealing for 250 °C (10 min) -Sulfurized at 525 °C (intermediate 200 °C, 250 °C and 300 °C) (180 s)	- Monoclinic - Large grains and homogeneous, - $E_g = 0.98\text{--}1.07$ eV	2026	This Study

The following are some of the solvents listed and their abbreviations; distilled water (DW), dimethyl sulfoxide (DMSO), EG (ethylene glycol), dimethylformamide (DMF), hydrochloric acid (HCl), monoethanolamine (MEA), 2-methoxyethanol (2ME)

structural and morphological characteristics of CTS thin films derived from both solution types.

2 Experimental details

2.1 Growth of CTS thin films

CTS precursor layers were coated on glass substrates by spin-coating method. Prior to deposition, the glass substrates were cut into 15 mm × 15 mm pieces and subjected to a standard cleaning process involving sequential 10 min ultrasonic cleaning steps in acetone, isopropanol (IPA), and distilled water (DW), followed by drying with nitrogen (N_2) gas. The precursor solution used in spin-coating consisted of Copper (II) Chloride dihydrate ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ —1 mmol/l); Tin (II) Chloride dihydrate ($\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ —0.5 mmol/l) and thiourea (H_2NCSNH_2 —4 mmol/l), and the precursors were dissolved separately in 10 mL of two different solvent systems: 2 methoxyethanol (2 ME) and a mixed solvent of 2-ME/DW (4:1 volume ratio). The prepared solutions were magnetically stirred at 50 °C for 2 h until they became clear and homogeneous. Subsequently, five drops of monoethanolamine (MEA) were added dropwise to each solution as a stabilizing agent. The solutions were stirred using a magnetic stirrer at room temperature for approximately 12 h, with the lid closed. They were spin-coated onto substrates using a dynamic method, as shown in Fig. 1. This procedure involved a two-step process: first, a low-speed spin at 500 rpm for 20 s, followed by a high-speed spin at 2500 rpm for 50 s to complete the coating. To

eliminate organic solvent residues, a drying process was performed on a hot plate at 250 °C for 10 min. This process was repeated several times until the thin film reached a targeted thickness of approximately 1.05–1.3 μm , as measured using a Dektak profilometer (Dektak XT).

After coating precursor films, the films were placed in a closed graphite box containing 10 mg of high purity sulfur powder for the sulfurization process in the rapid thermal processing (RTP) system. The precursor films were sulfurized using a temperature profile that included pre-annealing treatment for 3 min holding steps at 200 °C, 250 °C, and 300 °C, before reaching a reaction temperature of 525 °C in an Ar (95%) + H_2 (5%) atmosphere using a heating rate of 1 °C/s. To provide a basis for comparison, CTS thin films were prepared by sulfurization at 525 °C for 3 min without pre-annealing treatment. The experimental procedures were summarized in Table 2, and the annealing temperature profile is illustrated in Fig. 2. As seen in the table, the samples were coded into two types (Type-I and Type-II) through their production process. For example, the sample produced by sulfurizing precursors coated by using a pure 2-ME solution at 525 °C without pre-annealing, was denoted CTS-I. Conversely, the sample grown using an aqueous solution was labeled as CTS-II. The values at the end of the sample codes represent the pre-annealing temperature.

The solar cell was fabricated using the following sequence of processes. First, the CTS-I-200 absorber layer was deposited onto Mo-coated glass substrates. An n-type CdS buffer layer with a thickness

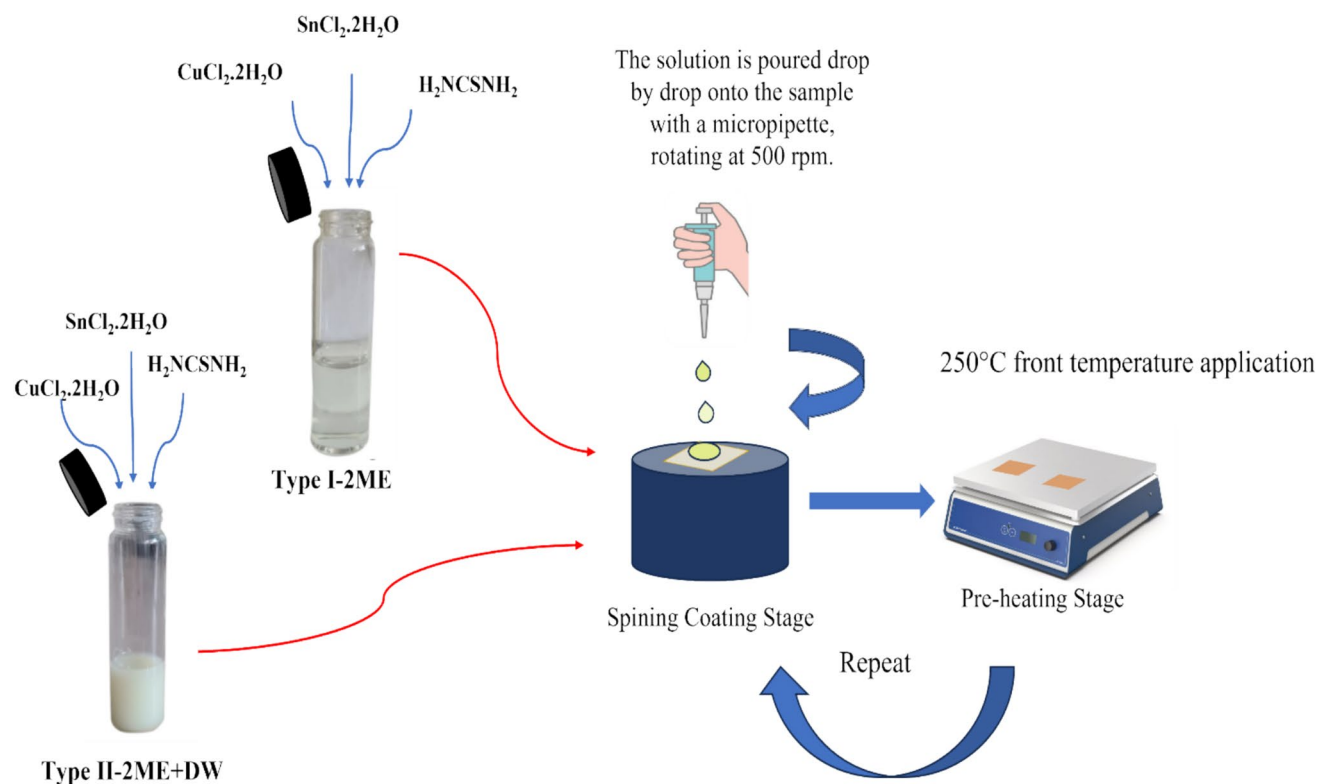


Fig. 1 Flow chart of the preparation process of CTS thin films obtained from different solvents produced by spin-coating method

Table 2 Prepared CTS thin films by employing different solvents and pre-annealing temperatures

	Samples	Solution (Ratio)	Pre-annealing	Sulfurization
Type 1	CTS-I	2-ME	-	525 °C, 3 min
	CTS-I-200	2-ME	200 °C, 3 min	
	CTS-I-250	2-ME	250 °C, 3 min	
	CTS-I-300	2-ME	300 °C, 3 min	
Type 2	CTS-II	2-ME/DW (4:1)	-	
	CTS-II-200	2-ME/DW (4:1)	200 °C, 3 min	
	CTS-II-200	2-ME/DW (4:1)	250 °C, 3 min	
	CTS-II-250	2-ME/DW (4:1)	300 °C, 3 min	

of approximately 60 nm was then deposited onto the CTS layer by chemical bath deposition (CBD), forming the glass/Mo/CTS/CdS structure. Subsequently, an

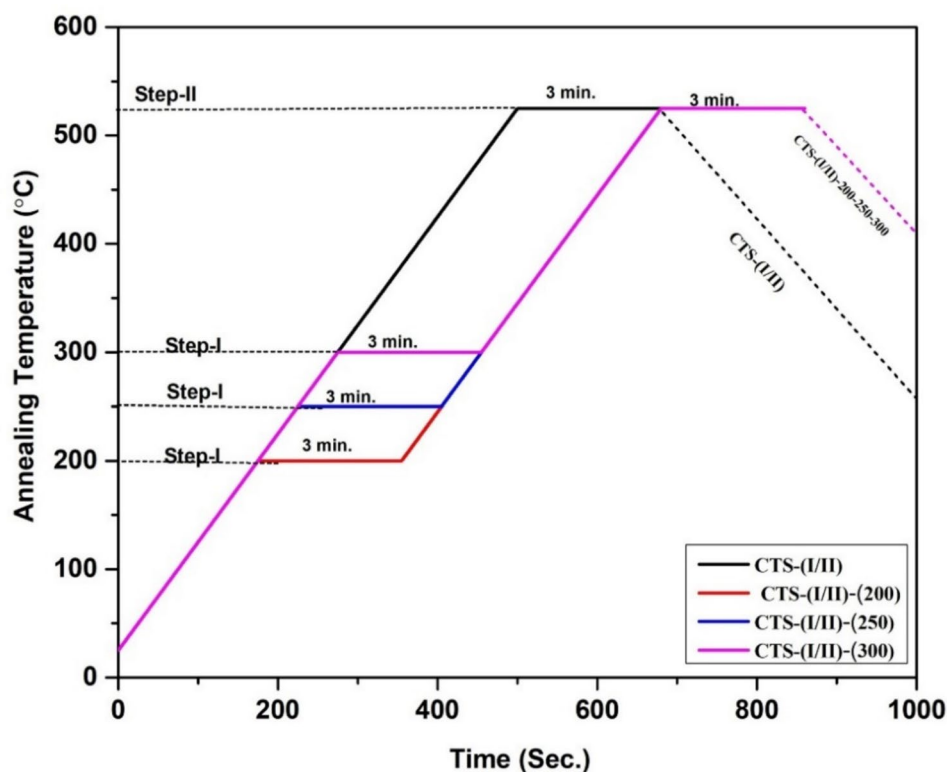
80 nm thick ZnO layer and a 300 nm thick Al-doped ZnO (AZO) transparent conducting layer were deposited by RF sputtering under optimized growth conditions. Finally, silver (Ag) top contacts were deposited to complete the glass/Mo/CTS-I-200/CdS/ZnO/AZO/Ag solar cell structure.

2.2 Characterization

The crystal structure, phase purity and the other structural properties of the samples were analyzed by using an X-ray diffractometer (using Cu-K α , $\lambda = 0.15406$ nm) and a Raman spectrometer. The formation mechanism of the CTS compound was investigated using Fourier Transform Infrared (FTIR) spectroscopy analysis performed with a VERTEX FT-IR/4100 spectrophotometer. The surface of the produced CTS films was examined using a scanning electron microscope (SEM, HITACHI SU5000) to investigate the effects of solution type and the pre-annealing treatment on the morphological properties of the films.

The atomic concentrations and compositions of CTS samples were determined by Energy Dispersive X-ray Spectroscopy (EDX). The optical transmittance of the

Fig. 2 Flow chart of the preparation process of CTS thin films obtained from different solvents



CTS films was determined by Spectroscopic Ellipsometry (J.A. Woollam V-VASE ellipsometer), while the basic electrical parameters such as resistivity and carrier concentration were determined using the Van der Pauw (Keithley 2410 source-meter) method under an approximate magnetic field density of 0.48 Tesla and a current of 0.1 mA through Hall Effect measurements. Furthermore, the current density–voltage (J–V) characteristics of the fabricated solar cell were measured under simulated AM 1.5G illumination with an intensity of 100 mW cm^{-2} using a calibrated solar simulator, with an active device area of approximately 1 cm^2 .

3 Results and discussion

3.1 Crystal structure properties

The XRD patterns of the CTS thin films prepared employing different solvents and pre-annealing temperatures are presented in Fig. 3. Regardless of the solution type and the pre-annealing temperature, a peak with preferred orientation appeared around $2\theta = \sim 28.4^\circ$ in all samples corresponding to the preferential (13–1) diffraction plane of the monoclinic CTS phase, which is preferred for solar cell applications

(JCPDS 98–009-1762). The other characteristic peaks associated with the monoclinic phase were also observed at approximately 2θ values of 16.1° , 32.9° , 47.3° and 56.1° corresponding to the (110), (131), (33–1), (33–3) planes in all samples. On the other hand, XRD analysis revealed peaks at $2\theta = \sim 26.9^\circ$ and $\sim 31.8^\circ$ corresponding to the CuS secondary phase (JCPDS 06–0464), which is expected due to the Cu-rich composition ($\text{Cu/Sn} > 2$) of the samples confirmed by EDX measurements (see Fig. 7). A variation in the amount of CuS phase formation was observed depending on the Cu content in the samples produced using the 2-ME solvent. This result is supported by the absence of the peak at 31.8° related to this phase in the CTS-I sample, because this sample has the lowest Cu/Sn ratio (2.13) among them. In the CTS-I sample, the formation of the SnS_2 phase (JCPDS 00–023-0677) was observed at $2\theta = 15.0^\circ$ in addition to the CuS phase. When the pre-annealing process was applied at 200°C , the SnS_2 phase disappeared, but the amount of CuS phase increased. It was found that increasing the pre-annealing temperature above 200°C led to the appearance of the Sn_2S_3 phase (JCPDS 01-075-2183) along with the CuS phase in the structure. In contrast, this was not the case for aqueous solution samples. It is noteworthy that despite having a higher Cu/Sn

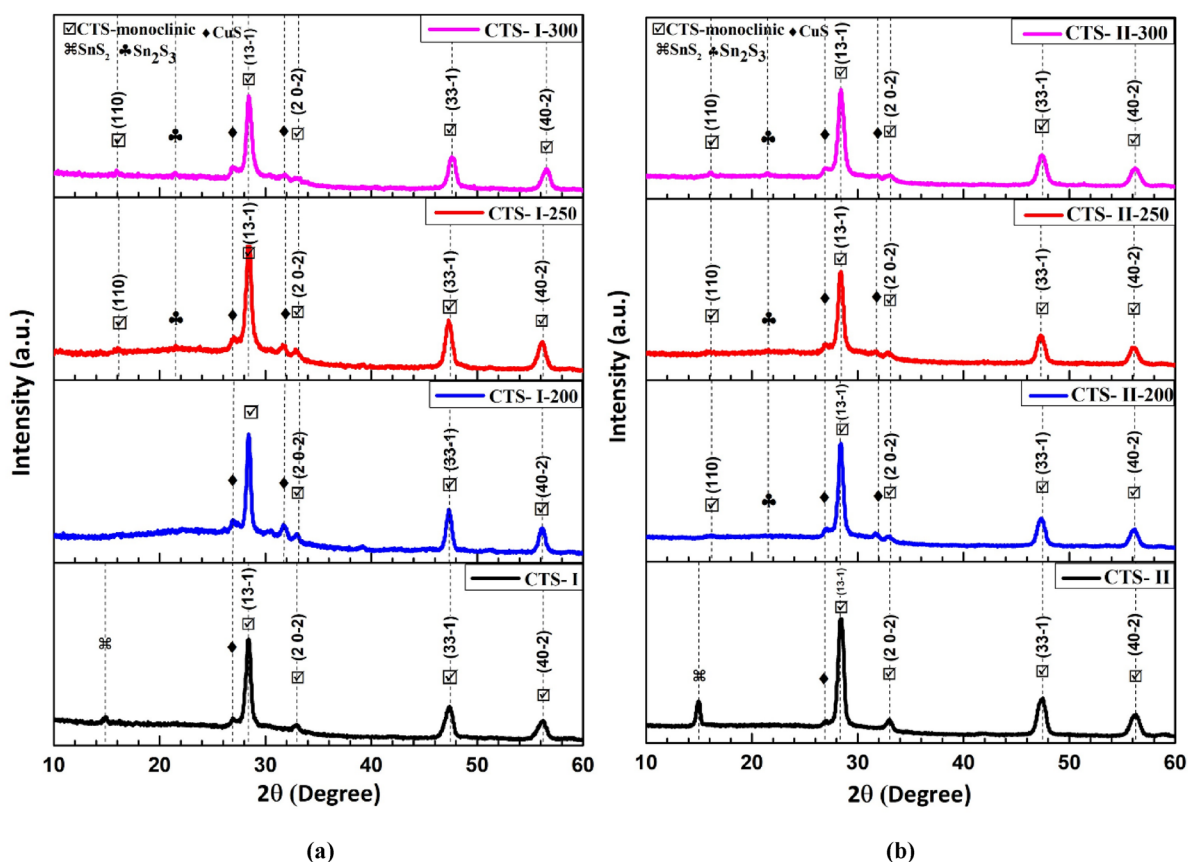


Fig. 3 XRD patterns of CTS thin films, Type-I (a) and Type-II (b)

ratio, the CTS-II sample shows a similar CuS phase formation to CTS-I. However, a dominant SnS_2 phase was observed at $2\theta = 15.0^\circ$, which should not be overlooked. The formation reaction for the CTS compound is $\text{Cu}_2\text{S} + \text{SnS}_2 \rightarrow \text{Cu}_2\text{SnS}_3$ [34]. However, it is thought that in aqueous solutions, the reaction between Sn and S, being more dominant than that between Cu and S, hinders the crystallization of the CuS phase, which is supported and discussed in detail by FTIR results.

On the other hand, pre-annealing treatment demonstrably affects the phase formation and distribution across both Type-I and Type-II sample sets. The pre-annealing treatment was found to lessen the deviation from the pure monoclinic CTS phase in CTS-I and CTS-II samples. Crucially, the SnS_2 phase formation, characterized by its volatility due to its high vapor pressure, was eliminated in the samples produced at 200°C . Nevertheless, upon applying a temperature of 250°C to the samples prepared in both sample sets, this phase re-emerges, having transformed into the Sn_2S_3 phase which was clearly seen in the samples treated at 300°C . Therefore, it was concluded that

pre-annealing at moderate temperatures (200°C) is more appropriate with respect to secondary phase-free CTS films.

To identify the most promising CTS film, some structural parameters were calculated and compared. The crystallite size of the CTS phase was determined by applying the Debye–Scherrer equation based on the most prominent peak of the (13–1) diffraction plane observed in the X-ray diffraction pattern. The average crystallite size (D) of the samples was calculated using Eq. 1 [35].

$$D = \frac{K\lambda}{\beta \cos \theta} \quad (1)$$

Here, β (in radians) represents the broadening of the diffraction peak measured at half maximum intensity (FWHM), while λ is the wavelength of the X-ray source radiation ($\text{Cu K}\alpha = 1.5406 \text{ \AA}$), and θ is the Bragg angle, and K is the Scherrer constant. The shape factor of the crystallites, denoted as K , is assumed to be 0.94 for spherical grain structures, as observed in SEM images. Equations 2 and 3 were used to calculate other

structural parameters, including microstrain (ϵ) and dislocation density (δ) [36]. Furthermore, the number of crystals per unit surface area (N) was determined using Eq. 4, where T represents the thickness of the film. The average crystallite size, dislocation density, and microstrain values for the CTS samples, obtained through these equations, are presented in Table 3.

$$\epsilon = \frac{\beta \cos \theta}{4} \quad (2)$$

$$\delta = \frac{1}{D^2} \quad (3)$$

$$N = \frac{T}{D^3} \quad (4)$$

The crystallite size attains maximum values of 22.52 nm and 21.39 nm for the CTS-I-200 and CTS-II-200 samples, respectively. Consistent with these findings, these samples exhibit the lowest strain and dislocation density (see Table 3). In contrast, the higher values observed in the sample sets annealed at 250 °C and 300 °C indicate structural distortion, and this is further supported by the lower crystallite size values. Upon comparison of Type-I and Type-II CTS samples, it can be inferred that samples fabricated using pure 2-ME demonstrate enhanced crystalline properties.

As summarized in Table 3, the largest crystallite size and the lowest microstrain, dislocation density, and number of crystals per unit surface area were observed in the CTS-I-200 sample, which was prepared using 2-ME and a pre-annealing temperature of 200 °C. These structural features namely, larger crystallite size and reduced microstrain, dislocation density, and grain boundary density, are advantageous for photovoltaic applications, as they contribute to improved

charge carrier mobility and reduced recombination losses, ultimately enhancing device performance. Overall, it was found that an intermediate pre-annealing temperature of 200 °C represents the optimal condition for both solution types concerning crystal size and phase purity.

3.1.1 Phase analysis

The Raman spectra of CTS samples are shown in Fig. 4, allowing a comparison of solvent types and pre-annealing temperatures. Peaks observed at around 296 cm^{-1} , 314 cm^{-1} , 352 cm^{-1} and 371 cm^{-1} correspond to the monoclinic phase in both Type-I and Type-II CTS samples which are consistent with the reported reports [37, 38]. Besides, a peak observed at 221 cm^{-1} belongs to the SnS phase in both types of films and this value is compatible with the SnS Raman data reported in the literature [39]. In the sample prepared with the aqueous solution at a pre-annealing temperature of 300 °C (CTS-II-300), a SnS peak at 182 cm^{-1} was observed as a secondary phase with minimal intensity [40]. However, it is interesting to note that while XRD analyses indicated the presence of a CuS phase potentially due to Cu-rich composition, no trace of it was detected in Raman spectroscopy. Raman spectra acquired from multiple measurements failed to confirm the presence of a CuS phase. This indicates that the CuS phase forms predominantly in the bulk regions of the samples rather than near the surface, because the penetration depth of the Raman laser source is more limited compared to that of X-rays [41]. The Raman peak intensity ratio, I_{296}/I_{352} was calculated, and it was found to be below unity (0.96–0.99) for samples synthesized using pure 2-ME. Conversely, for samples synthesized using the (2-ME + DW)

Table 3 Some structural parameters of the CTS samples prepared employing different pre-annealing temperatures

	Samples	FWHM (β°)	D (nm)	ϵ (10^{-3}) (a.u.)	δ (10^{-3} nm^{-2})	N (nm^{-2})
Type-I	CTS-I	0.39	21.94	1.64	2.07	12.29
	CTS-I-200	0.38	22.52	1.60	1.97	11.37
	CTS-I-250	0.48	17.83	2.00	3.14	22.92
	CTS-I-300	0.49	17.46	2.07	3.27	24.39
	CTS-II	0.53	16.14	2.24	3.83	30.86
Type-II	CTS-II-200	0.40	21.39	1.69	2.18	13.27
	CTS-II-250	0.46	18.60	1.94	2.88	20.18
	CTS-II-300	0.48	17.83	2.03	3.14	22.93

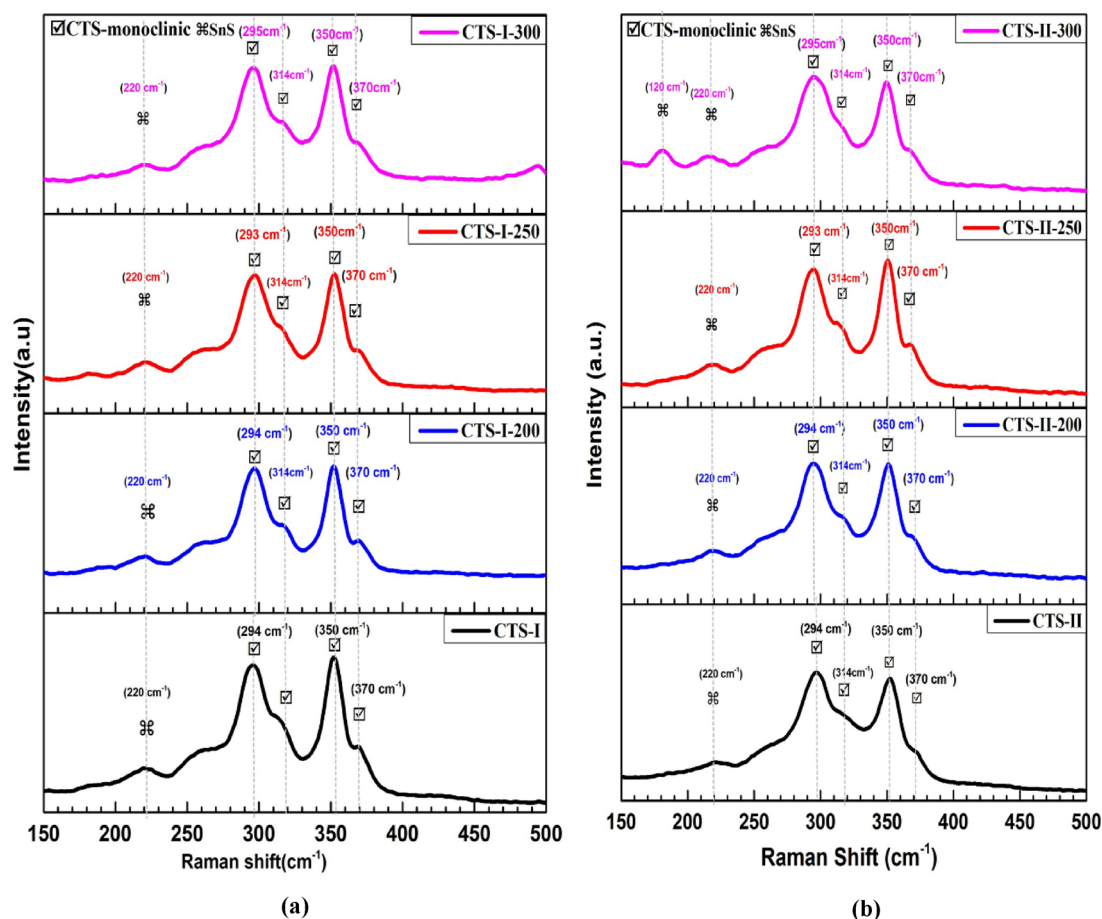


Fig. 4 Raman spectra of CTS thin films; **a** Type (I), **b** Type (II)

mixture solution, the ratio was determined to be above unity (1.00–1.06), with the exception of CTS-II-250. This finding implies that the choice of solvent influences the short-range ordering within the material. A study that can be conducted differently from the scope of this work has shown that for cubic CTS thin films, the intensities of the characteristic main peaks at 303 and 354 cm^{-1} exhibit a relative change as the Cu content in the CTS phase is increased. This change in peak intensity ratio is reported to demonstrate an alteration in short-range order of atoms [42].

Raman spectroscopy provides important insight into the effects of solvent chemistry and pre-annealing temperature on phase formation and local bonding in CTS thin films. The use of pure 2-ME results in spectra dominated by monoclinic CTS vibrational modes with suppressed secondary phase signatures, indicating improved chemical homogeneity and stronger Cu–S and Sn–S bonding within the CTS lattice. In contrast, the mixed (2-ME + DW) solvent promotes the appearance of

SnS-related Raman peaks, suggesting that water facilitates partial phase segregation, particularly at higher pre-annealing temperatures. Increasing the pre-annealing temperature enhances the intensity and sharpness of the characteristic CTS modes, reflecting improved crystallinity; however, excessive temperature in the aqueous system leads to the emergence of additional SnS features. The Raman intensity ratio of the main CTS modes further confirms solvent-dependent variations in short-range ordering. Considering minimized secondary phase formation and enhanced CTS bonding characteristics, films synthesized using pure 2-ME at moderate pre-annealing temperatures (200 °C) are identified as the optimal condition within the investigated parameter space.

3.1.2 Functional groups and chemical bonding

Figure 5 presents the FTIR spectra obtained in the 500–3000 cm^{-1} wavelength range for the CTS-I and CTS-II samples. The measurements were carried out using powders obtained by scraping thin films deposited on glass substrates. The vibrations observed in the spectra correspond to similar functional groups. In this context, the sharp band observed at 610 cm^{-1} is attributed to the Cu–S stretching vibration, which confirms the presence of the CuS secondary phase, as also observed in the XRD analysis. On the other hand, in the CTS-I sample, the bands at 1395 and 2342 cm^{-1} , which correspond to the Sn–S stretching vibration, support partial phase separation [43, 44]. In the CTS-II sample, the increased intensity of these bands, along with the emergence of a new band at 852 cm^{-1} attributed to Sn–S, indicates a more pronounced presence of SnS. This situation can be explained as follows. MEA's amine group (NH_2) is a strong nucleophile, capable of directly coordinating with Sn^{2+} . Beyond simple hydrolysis, MEA can function as a ligand, primarily binding through its nitrogen atom, though coordination via the methoxy oxygen is also possible. Overall, the use of DW with MEA as a solvent for the SnCl_2

precursor altered the hydrolysis process, potentially complicating the control over the formation of tin-containing phases with enhanced stability and solubility in solvents. This observation is directly supported by the XRD results.

3.1.3 Morphological properties

SEM surface images of the samples are presented in Fig. 6, revealing a distinct effect of the solution type and pre-annealing temperature on the film morphology. Upon examination of Type-I CTS samples, the sample without any pre-annealing exhibited a structure consisting of large grains on the surface along with smaller grains beneath. EDXS measurements taken from the large grains indicated a Cu/Sn ratio exceeding 3, corroborating the phase separation (CuS) within the structure as detected by XRD.

In the CTS-I-200 sample, it was determined that the large grains associated with the CuS phase dispersed and evolved into smaller grains, with a partial improvement observed in the compactness of the structure. Furthermore, while the CTS-I-250 sample exhibited a more homogeneously distributed granular

Fig. 5 FTIR spectra of CTS-I and CTS-II samples

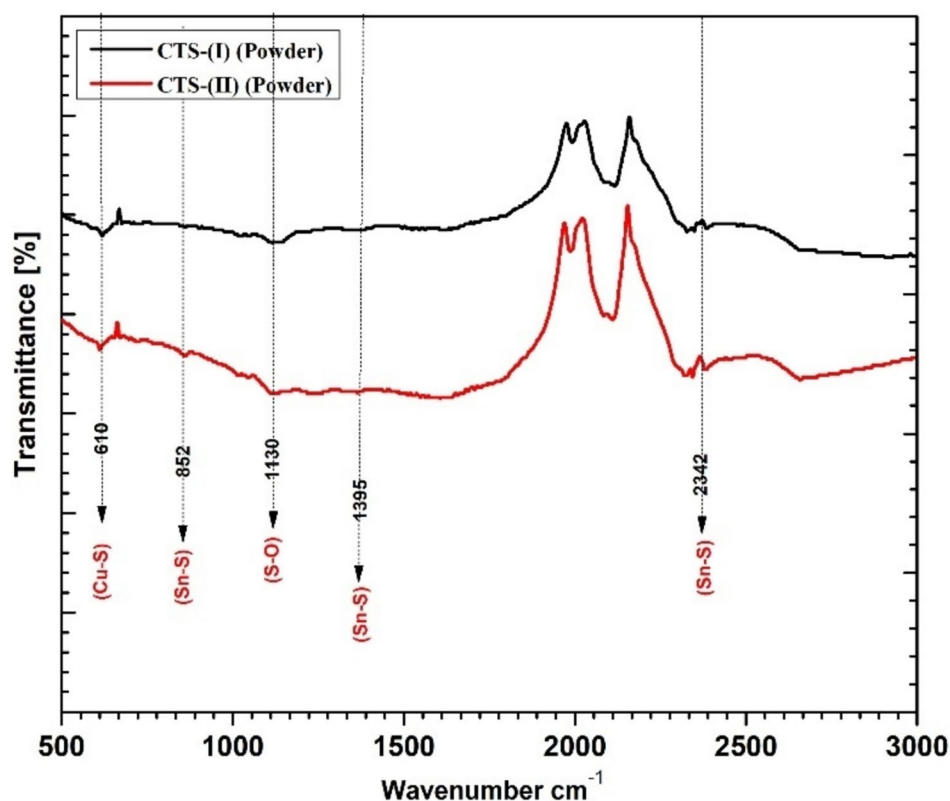
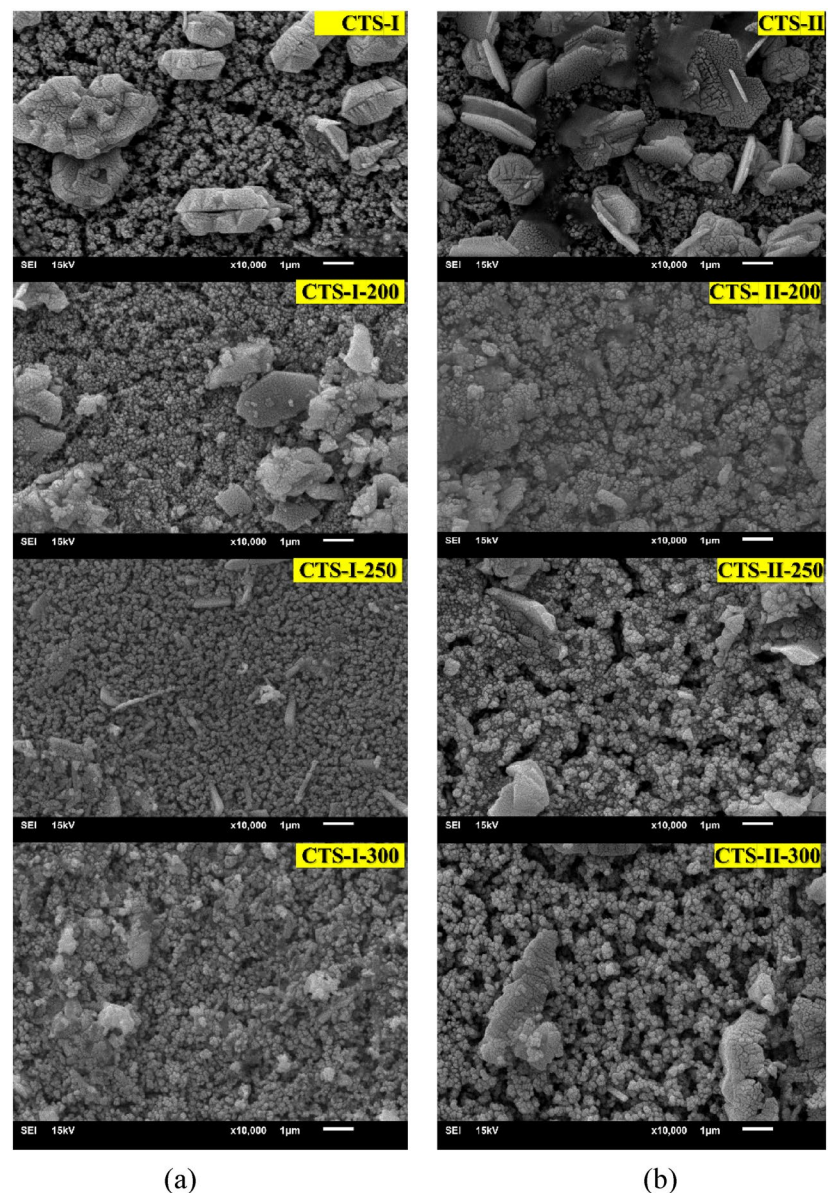


Fig. 6 High-magnification surface SEM images of CTS thin films; **a** Type (I), **b** Type (II)



structure, it was determined that embedded plate-like grain formations appeared within the fine-grained structure. This indicates that the interlayer temperature has a significant effect on the morphology of the thin films. The findings obtained from Raman and EDXS analyses suggest that these morphological variations can be attributed to the differences in the sulfur content and density of various tin sulfide phases such as SnS and Sn_2S_3 . In the CTS-I-300 sample, the grains within the structure were observed to coalesce, exhibiting a granular structure with indistinct boundaries and a nonhomogeneous distribution in a molten-like form.

On the other hand, in Type-II CTS thin films, while exhibiting a morphology similar to the CTS-I film, plate-like structures related to the dominant Sn-S phase appearing in the XRD spectrum were formed. However, in CTS-II-200, the formations related to these secondary phases almost completely disappeared reflecting improved morphology of the film. Larger grain sizes, which result in fewer grain boundaries, are generally associated with improved device performance [45]. It was observed that large grains, nonuniformly distributed, tend to form densely as embedded clusters within the finer layer. As the pre-annealing temperature increases, the initially compact

structure deteriorates, leading to a film morphology characterized by inhomogeneous distribution and the appearance of cracks in certain regions. As a result, when comparing the pre-annealing effects on the surface structures of both types of CTS films, the samples annealed at 200 °C exhibited more promising morphology compared to the others with respect to both homogeneity and larger grains. The high-magnification SEM

images (10 kX) reveal the presence of surface cracks and confirm the overall homogeneity of the CTS films (Fig. 6).

3.1.4 EDXS

Figure 7 presents the images of EDXS spectra, along with selected atomic ratios (in insets), of Type-I and

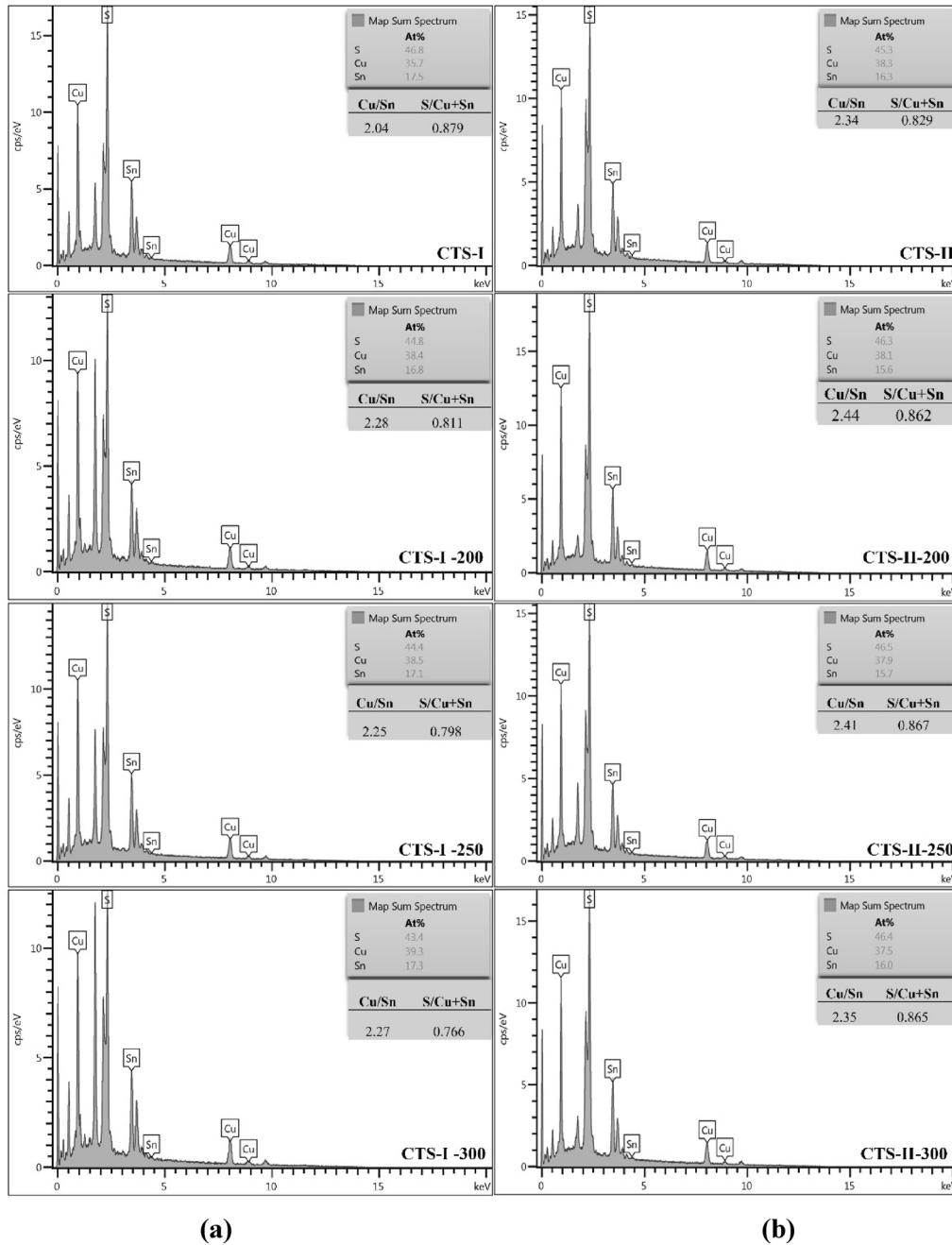


Fig. 7 EDX spectra of CTS samples; **a** Type- I and **b** Type-II

Type-II CTS thin films. Irrespective of the precursor solution employed in the synthesis, the samples exhibit a Cu-rich nature, characterized by a Cu/Sn ratio exceeding 2. While the Cu/Sn atomic ratio changed between 2.04 and 2.29 for Type-I CTS samples, it changed between 2.34 and 2.44 for Type-II CTS samples. This observation suggests the formation of a Cu-rich CuS phase, in addition to the monoclinic CTS phase identified in XRD spectra. Pre-annealing treatment does not significantly alter the material composition for both sample sets; however, Type-I CTS films demonstrate a comparatively less pronounced Cu-rich composition. It can be said that the partial loss of tin in these samples leads to a chemical composition rich in copper. Moreover, the elemental distributions in CTS-II-200, CTS-II-250 and CTS-II-300 were found to be more stoichiometric than the nonaqueous solutions and the Cu/Sn ratios of the samples made with nonaqueous solutions were lower. To produce high-quality PV thin-film absorber layers, precise control of the Cu/Sn ratio, which is one of the fundamental compositional parameters of the CTS material, is essential.

It is interesting to note that, as the pre-annealing temperature increased for Type-I CTS samples, the sulfur content decreased from 0.87 to 0.76 (see S/Metal ratio). On the other hand, the S/Metal ratio increased from 0.82 to 0.86 for Type-II CTS samples.

This behavior might be attributed to the nature of the solvent used in the solution preparation. In Type-II CTS samples, where an aqueous solution is employed, sulfur atoms are more effectively retained during the pre-annealing process due to the stronger interaction between sulfur species and the polar solvent. This leads to a gradual increase in the S/Metal ratio with temperature. In contrast, the organic solvent used in Type-I samples may result in higher sulfur loss at elevated temperatures, leading to a decrease in sulfur content.

3.1.5 Optical properties

The absorption coefficients of the samples were calculated using the transmittance values measured in the wavelength range of 600–1350 nm at room temperature. According to the Lambert–Beer law [46], the absorption coefficient is determined by Eq. 6.

$$a = \frac{1}{d} \ln\left(\frac{1}{T}\right) \quad (6)$$

In this equation, α represents the absorption coefficient, d denotes the thickness of the sample, and T stands for the optical transmittance. The determination of the optical bandgap (E_g) of CTS thin films with varying thicknesses was conducted at different annealing temperatures based on the model proposed by Tauc in Eq. 7 [47].

$$(\alpha h\nu) = A(h\nu - E_g)^{\frac{1}{2}} \quad (7)$$

From these graphs, the optical bandgap (E_g) of the CTS film was determined by extrapolating the linear portion of the curve to the x-axis. The band structure of Cu-based ternary, quaternary, or pentanary compounds can be related to the degree of crystallinity and phase purity, as well as being directly influenced by the amount of Cu present in the structure. The optical bandgaps for both types of CTS samples were extracted from the data presented in Fig. 8. For Type-I CTS thin films, the intermediate annealing process has a limited impact on the optical bandgap, with E_g values varying within a narrow range of 0.98–1.02 eV. In contrast, for Type-II CTS thin films, the same annealing treatment significantly influences the optical bandgap, resulting in a broader variation from 0.99 to 1.07 eV. This indicates that Type-II films are more sensitive to thermal treatment in terms of their optoelectronic properties. Given that the strength of the interaction between Cu and S orbitals is inversely proportional to E_g , the observation that the CTS-I-200 sample, with the highest Cu/Sn ratio, exhibits the lowest E_g aligns well with previous reports [48]. Similarly, the analysis results for Type-II CTS indicate that the Cu/Sn ratio is highest in the CTS-II-200 sample, while E_g is the lowest.

The determination of the bandgap value is significantly influenced not only by the Cu content but also by the presence of secondary phases, with a trade-off existing between the two factors. Conversely, in Type-II CTS samples, the extent of the pre-annealing effect is comparatively more significant. Although the Cu/Sn ratio is higher in the CTS-II sample compared to CTS-I, the higher bandgap value can be attributed to the more pronounced SnS_2 (approximately 2.2 eV) secondary phase with a high bandgap in the CTS-II sample, which is observed in XRD. On the other hand, it can be stated that in Type-II samples, the variations in the bandgap depend on the Cu content.

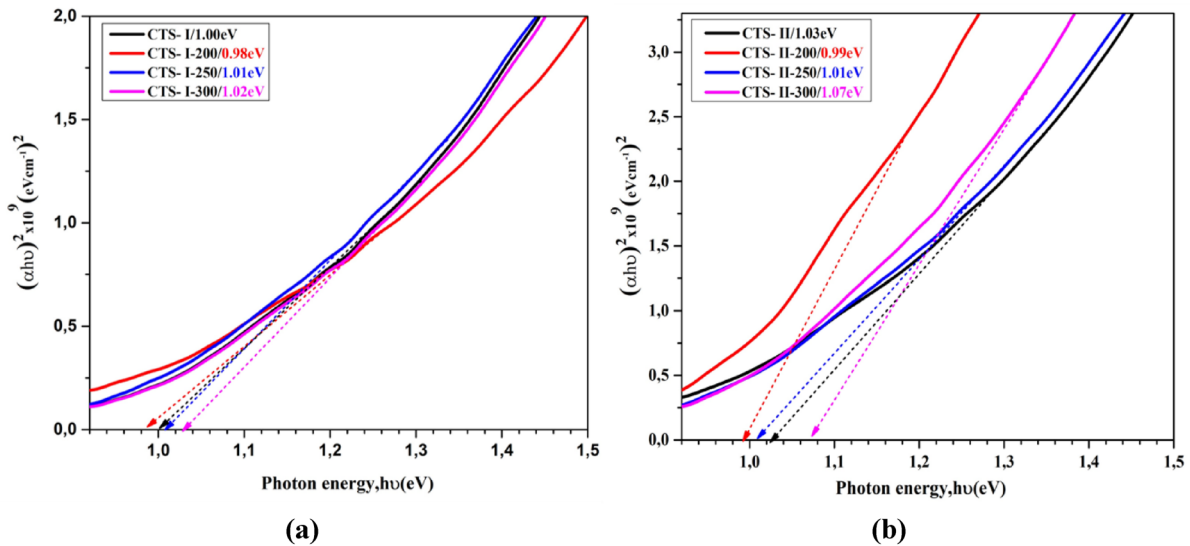


Fig. 8 $(\alpha h\nu)^2$ versus photon energy ($h\nu$) plots for the determination of energy bandgap **a** Type-I **b** Type-II CTS samples

Table 4 Resistivity and carrier density values of the Type-I and Type-II CTS samples

	Sample	Carrier type	Resistivity (Ohm-cm)	Carrier concentration (cm^{-3})
Type-I	CTS-I	p	4.25	1.08×10^{17}
	CTS-I-200		3.96	1.28×10^{17}
	CTS-I-250		3.01	2.38×10^{17}
	CTS-I-300		4.78	2.80×10^{16}
Type-II	CTS-II	p	1.64	5.86×10^{17}
	CTS-II-200		1.54	6.59×10^{17}
	CTS-II-250		1.35	3.09×10^{17}
	CTS-II-300		1.67	9.27×10^{17}

3.1.6 Electrical properties

The electrical properties such as resistivity, carrier density, and carrier type of the CTS samples, determined by the Van der Pauw method, are presented in Table 4. While both types of CTS films exhibit p-type conductivity, annealing and solvent type have an effect on the other electrical parameters. Type-II CTS thin films showed higher carrier concentration and lower resistivity compared to Type-I samples. When comparing the carrier concentration of CTS films prepared with 2-ME versus those prepared with 2-ME + DW, the 2-ME solution provides lower polarity, resulting in more controlled and slower crystal growth. This may lead to the formation of

more uniform crystals with fewer defects, which consequently generate fewer free holes and reduce the carrier density. In contrast, deionized water is a highly polar solvent that may cause rapid nucleation during crystal growth. This rapid growth typically produces more defects and creates p-type carriers, resulting in higher carrier density [49].

This effect can be attributed to the presence of CuS, which is a well-known p-type semiconductor. Due to its degenerate or heavily doped nature, CuS significantly contributes to the overall increase in hole concentration within the CTS matrix [50]. Additionally, the lowest resistivity observed in the CTS-I-250 and CTS-II-250 films can be associated with their better surface morphology. Overall, it can be concluded that the solution type of precursor films and the pre-annealing treatment had a great impact on the electrical properties of CTS films.

The superior optoelectronic behavior observed for the CTS-I-200 film can be directly attributed to its stabilization in the monoclinic CTS phase, which is associated with a purer crystal structure and enhanced structural order. The monoclinic phase promotes a more ordered Cu–Sn cation arrangement, suppressing antisite disorder and defect formation that are commonly present in cubic and tetragonal CTS phases. This increased structural ordering facilitates enhanced grain growth, resulting in a larger crystallite size, while simultaneously reducing microstrain and dislocation density within

the lattice. The reduced density of structural defects and grain boundaries minimizes carrier scattering and non-radiative recombination pathways, leading to improved charge transport and more stable electrical behavior. Consequently, the combination of monoclinic phase purity, larger crystallite size, and lower microstrain and dislocation density in the CTS-I-200 sample establishes a clear processing–structure–property relationship that underpins its superior optical absorption characteristics and electrical performance, highlighting its suitability as an absorber layer for photovoltaic applications.

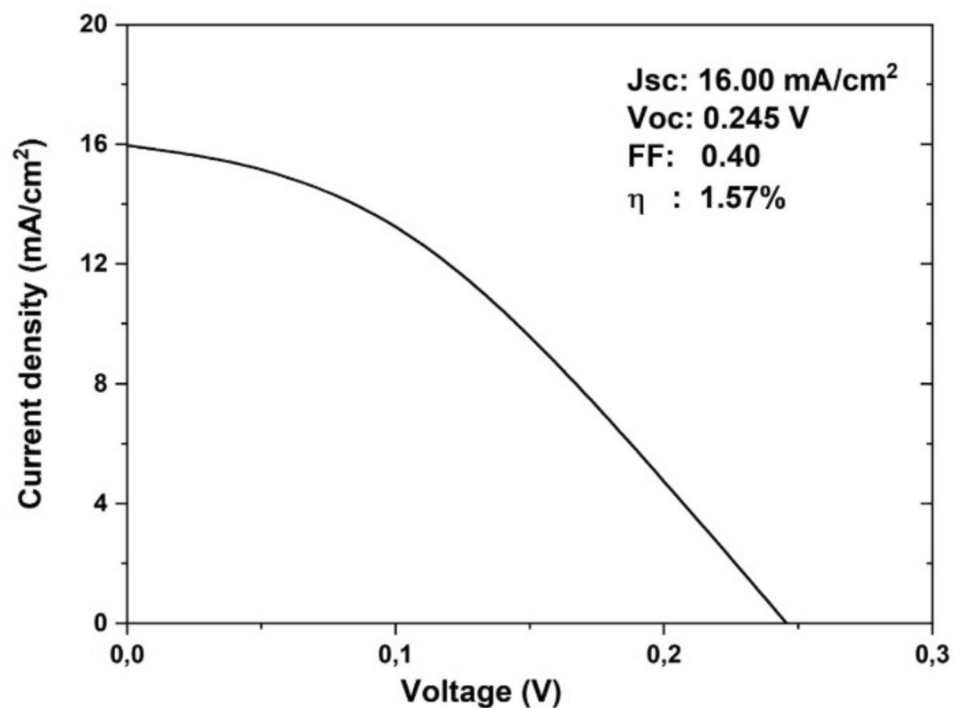
3.1.7 Photovoltaic properties

To evaluate the photovoltaic performance of the chemically prepared CTS-I-200 absorber layer, a device with the structure Mo/CTS-I-200/CdS/ZnO/AZO/Ag was fabricated. The photovoltaic characteristics of the solar cell were assessed by current density–voltage (J–V) measurements under standard illumination conditions, as shown in Fig. 9. The device exhibited an open-circuit voltage (V_{oc}) of 0.245 V, a short-circuit current density (J_{sc}) of 16.00 mA cm^{-2} , a fill factor (FF) of 0.40, and a corresponding power conversion efficiency (PCE) of approximately 1.57%.

CTS films processed using 2-ME exhibit superior structural quality compared to those prepared with 2-ME/DW, primarily due to more controlled precursor decomposition. Moderate annealing temperatures ($200 \text{ }^{\circ}\text{C} \leq$) provide sufficient atomic mobility for Cu and Sn cations to rearrange into the thermodynamically stable monoclinic phase, thereby enhancing crystallinity and phase purity. This promotes grain growth while reducing microstrain and dislocation density, and effectively suppresses secondary and metastable phases without inducing excessive sulfur loss. At higher pre-annealing temperatures, grain size decreases due to enhanced surface diffusion and grain boundary migration, which can increase defect density. In addition, the optical bandgap increases with annealing temperature, leading to reduced optical absorption. Consequently, moderate annealing temperatures offer an optimal balance, resulting in improved stoichiometry, reduced defect density, and enhanced overall film quality.

From a theoretical standpoint, processing parameters, particularly solvent composition and annealing temperature, govern the nucleation and growth kinetics, precursor decomposition, elemental diffusion, and sulfur volatilization during CTS thin-film formation. These processes determine elemental distribution, phase purity, crystal structure, grain size,

Fig. 9 J–V measurement of the Mo/CTS-I-200/CdS/ZnO/AZO/Ag cell



and grain boundary density, which are critical factors controlling defect density and carrier transport. In the context of solar cell operation, improved crystallinity and reduced grain boundary density suppress non-radiative recombination pathways, thereby enhancing minority carrier lifetime and diffusion length within the absorber layer. Furthermore, controlled sulfur retention and phase stability prevent the formation of high-bandgap or electrically inactive secondary phases that can act as recombination centers or potential barriers at grain boundaries. The resulting optimization of bandgap energy, carrier concentration, and electrical resistivity directly influences key photovoltaic parameters, including open-circuit voltage, short-circuit current density, and fill factor. Therefore, precise control of solvent chemistry and thermal processing is essential for maximizing the photovoltaic performance of CTS-based solar cells through improved absorber quality and charge transport efficiency.

4 Conclusion

In conclusion, this study demonstrates that both precursor solution chemistry (2-ME and 2-ME/DW) and pre-annealing temperature play a decisive role in tailoring the structural, optical, and electrical properties of Cu_2SnS_3 thin films. The solvent type strongly influences precursor decomposition, crystallization behavior, sulfur retention, secondary phase and charge transport. Among all samples, the CTS-I-200 film, prepared using 2-ME and pre-annealed at 200 °C, exhibited the most favorable characteristics, including the largest crystallite size (22.52 nm) and the lowest micro-strain (1.6×10^{-3}), dislocation density ($1.97 \times 10^{-3} \text{ nm}^{-2}$), and grain boundary density (1.14 nm^{-2}). These features are beneficial for photovoltaic applications as they promote efficient charge transport and reduced recombination.

Raman and XRD analyses confirmed the formation of the monoclinic CTS phase and indicated that CuS secondary phases are predominantly present in the bulk rather than on the surface, consistent with the different penetration depths of X-rays and Raman excitation. Elemental analysis revealed a Cu-rich composition for all films. However, sulfur retention exhibited a strong dependence on solvent chemistry: sulfur content decreased with increasing temperature in Type-I (2-ME-based) films, whereas it increased in Type-II

(aqueous-based) films, indicating enhanced sulfur retention in the presence of distilled water.

Optical measurements showed that the bandgap varied within a narrow range for Type-I films (0.98–1.02 eV) and over a wider range for Type-II films (0.99–1.07 eV), reflecting greater sensitivity to annealing in aqueous systems and the influence of the Cu/Sn ratio and secondary phases such as SnS_2 . Electrical characterization confirmed p-type conductivity for all samples. The CTS-I-200 film exhibited a carrier concentration of $1.28 \times 10^{17} \text{ cm}^{-3}$ and a resistivity of $3.96 \text{ } \Omega\text{-cm}$, while Type-II films generally showed higher carrier concentrations and lower resistivities due to increased Cu-rich secondary phase content.

Overall, these results highlight the critical importance of coordinated control over solution composition and pre-annealing treatment in optimizing CTS thin films. Based on the combined structural, optical, and electrical analyses, the CTS-I-200 sample emerges as the most promising absorber candidate for CTS-based photovoltaic device development, as evidenced by a cell efficiency of 1.57%.

Acknowledgements

The authors gratefully acknowledge funding from The Scientific and Technological Research Council of Türkiye (TUBITAK) under project number 122F217.

Author contributions

S. Erkan: Sample preparation, Characterization, Writing—original draft. Y. Atasoy: Characterization, Writing—original draft. M.A. Olgar: Methodology, Writing—review & editing. R. Zan: Conceptualization, Methodology, Writing—review & editing.

Funding

Open access funding provided by the Scientific and Technological Research Council of Türkiye (TUBITAK). The authors gratefully acknowledge funding from The Scientific and Technological Research Council of Türkiye (TUBITAK) under project number 122F217.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval This article does not contain any studies involving animals performed by any of the authors. Also, this article does not contain any studies involving human participants performed by any of the authors.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. M.J. De Wild-Scholten, Energy payback time and carbon footprint of commercial photovoltaic systems. *Sol. Energy Mater. Sol. Cells* **119**, 296–305 (2013). <https://doi.org/10.1016/j.solmat.2013.08.037>
2. L. Zhang, Z. Xu, A critical review of material flow, recycling technologies, challenges and future strategy for scattered metals from minerals to wastes. *J. Clean. Prod.* **202**, 1001–1025 (2018). <https://doi.org/10.1016/j.jclepro.2018.08.073>
3. I.M. El Radaf, A.S. Hassanien, Effect of thickness on structural, optical, and optoelectrical properties of sprayed CuInSnS₄ thin films as a new absorber layer for solar cells. *Physica B* (2023). <https://doi.org/10.1016/j.physb.2023.414867>
4. G.P. Reddy, K.T.R. Reddy, Preparation and characterization of Cu₂SnS₃ thin films by two stage process for solar cell application. *Mater. Today: Proc.* **4**(13), 12401–12406 (2017)
5. Y. Gong, A. Jimenez-Arguijo, I. Caño, R. Scaffidi, C. Malerba, M. Valentini et al., Attaining 15.1% efficiency in Cu₂ZnSnS₄ solar cells under indoor conditions through sodium and lithium codoping. *Solar RRL* (2025). <https://doi.org/10.1002/solr.202400756>
6. L. Ma, W. Shi, L.M. Woods, Structural diversity of CuZn₂InSe₄ quaternary chalcogenides: electronic and phonon properties from first principles. *RSC Adv.* **12**, 26648–26656 (2022). <https://doi.org/10.1039/d2ra04261d>
7. N.J. Choudhari, Y. Raviprakash, S.D. George, M.S. Murari, Effect of stacking sequence on the structural ordering and phase formation in the sequentially deposited copper zinc tin sulfide thin films. *Int. J. Energy Res.* **46**, 7758–7774 (2022). <https://doi.org/10.1002/er.7678>
8. S. Rahaman, M.K. Singha, M.A. Sunil, K. Ghosh, Effect of copper concentration on CTS thin films for solar cell absorber layer and photocatalysis applications. *Superlattices Microstruct.* (2020). <https://doi.org/10.1016/j.spmi.2020.106589>
9. A.C. Lokhande, R.B.V. Chalapathy, M. He, E. Jo, M. Gang, S.A. Pawar et al., Development of Cu₂SnS₃ (CTS) thin film solar cells by physical techniques: a status review. *Sol. Energy Mater. Sol. Cells* **153**, 84–107 (2016). <https://doi.org/10.1016/j.solmat.2016.04.003>
10. H.D. Shelke, A.C. Lokhande, J.H. Kim, C.D. Lokhande, Photoelectrochemical (PEC) studies on Cu₂SnS₃ (CTS) thin films deposited by chemical bath deposition method. *J. Colloid Interface Sci.* **506**, 144–153 (2017). <https://doi.org/10.1016/j.jcis.2017.07.032>
11. I.Y. Kim, J.Y. Lee, U.V. Ghorpade, M.P. Suryawanshi, D.S. Lee, J.H. Kim, Influence of annealing temperature on the properties and solar cell performance of Cu₂SnS₃(CTS) thin film prepared using sputtering method. *J. Alloys Compd.* **688**, 12–17 (2016). <https://doi.org/10.1016/j.jallcom.2016.06.264>
12. Z. Jia, Q. Chen, J. Chen, T. Wang, Z. Li, X. Dou, The photovoltaic properties of novel narrow band gap Cu₂SnS₃ films prepared by a spray pyrolysis method. *RSC Adv.* **5**, 28885–28891 (2015). <https://doi.org/10.1039/c5ra01610j>
13. H.D. Shelke, A.C. Lokhande, V.S. Raut, A.M. Patil, J.H. Kim, C.D. Lokhande, Facile synthesis of Cu₂SnS₃ thin films grown by SILAR method: effect of film thickness. *J.*

- Mater. Sci. Mater. Electron. **28**, 7912–7921 (2017). <https://doi.org/10.1007/s10854-017-6492-7>
14. S.A. Vanalakar, G.L. Agawane, S.W. Shin, M.P. Suryawanshi, K.V. Gurav, K.S. Jeon et al., A review on pulsed laser deposited CZTS thin films for solar cell applications. *J. Alloys Compd.* **619**, 109–121 (2015). <https://doi.org/10.1016/j.jallcom.2014.09.018>
15. N.R. Mathews, J. Tamy Benítez, F. Paraguay-Delgado, M. Pal, L. Huerta, Formation of Cu₂SnS₃ thin film by the heat treatment of electrodeposited SnS-Cu layers. *J. Mater. Sci. Mater. Electron.* **24**, 4060–4067 (2013). <https://doi.org/10.1007/s10854-013-1361-5>
16. D.M. Berg, R. Djemour, L. Gütay, G. Zoppi, S. Siebentritt, P.J. Dale, Thin film solar cells based on the ternary compound Cu₂SnS₃. *Thin Solid Films* **520**, 6291–6294 (2012). <https://doi.org/10.1016/j.tsf.2012.05.085>
17. M.B. Zaman, V. Shrotriya, A. Hossain, I.M. Mehedi, M.M. Alam, Non hydrazine based chemical synthesis of earth abundant Cu₂SnS₃ thin film photocatalyst for wastewater treatment. *Ceram. Int.* **49**, 20822–20831 (2023). <https://doi.org/10.1016/j.ceramint.2023.03.215>
18. A. Cho, S. Banu, Y. Cho, S.K. Ahn, J.H. Yun, J.S. Cho, The effect of metal-chelate complex in Cu₂SnS₃ thin film solar cells and their characteristics, photovoltaic performance, and defect analysis. *Sol. Energy* **185**, 131–145 (2019). <https://doi.org/10.1016/j.solener.2019.04.065>
19. T. Bayazit, M.A. Olgar, T. Küçükömeroğlu, E. Bacaksız, M. Tomakin, Growth and characterization of Cu₂SnS₃ (CTS), Cu₂SnSe₃ (CTSe), and Cu₂Sn(S,Se)₃ (CTSSe) thin films using dip-coated Cu–Sn precursor. *J. Mater. Sci. Mater. Electron.* **30**, 12612–12618 (2019). <https://doi.org/10.1007/s10854-019-01622-4>
20. S. Dias, K. Kumawat, S. Biswas, S.B. Krupanidhi, Solvothermal synthesis of Cu₂SnS₃ quantum dots and their application in near-infrared photodetectors. *Inorg. Chem.* **56**, 2198–2203 (2017). <https://doi.org/10.1021/acs.inorgchem.6b02832>
21. M.H. Sayed, E.V.C. Robert, P.J. Dale, L. Gütay, Cu₂SnS₃ based thin film solar cells from chemical spray pyrolysis. *Thin Solid Films* **669**, 436–439 (2019). <https://doi.org/10.1016/j.tsf.2018.11.002>
22. M.P. Suryawanshi, U.V. Ghorpade, S.W. Shin, S.A. Pawar, I.Y. Kim, C.W. Hong et al., A simple aqueous precursor solution processing of earth-abundant Cu₂SnS₃ absorbers for thin-film solar cells. *ACS Appl. Mater. Interfaces* **8**, 11603–11614 (2016). <https://doi.org/10.1021/acsami.6b02167>
23. Q. Chen, D. Ma, Preparation of nanostructured Cu₂SnS₃ photocatalysts by solvothermal method. *Int. J. Photoenergy* (2013). <https://doi.org/10.1155/2013/593420>
24. Y.M. Ko, R.B.V. Chalapathy, L. Larina, B.T. Ahn, Growth of a void-free Cu₂SnS₃ thin film using a Cu/SnS₂ precursor through an intermediate-temperature pre-annealing and sulfurization process. *CrystEngComm* **19**, 5764–5773 (2017). <https://doi.org/10.1039/c7ce01261f>
25. M. Yang, X. Huang, L. Yao, L. Lin, A. Chen, Y. Chen et al., Self-stabilizing molecular solution for Cu₂SnS₃ thin film: an insight into the oxidation inhibitor of bivalent tin ion. *J. Power. Sources* (2021). <https://doi.org/10.1016/j.jpowsour.2021.229699>
26. M.K. Gadha, T.V. Anitha, P. Punathil, T.V. Vimalkumar, Annealing temperature optimization for dip-coated Cu₂SnS₃ thin films: sustainable pathway to CTS/Zn(O, S) solar cells via numerical simulation. *Mater. Sci. Eng. B* (2025). <https://doi.org/10.1016/j.mseb.2025.118003>
27. S.M. Bhise, H.D. Shelke, A. Al-Ahmed, M.A. Patil, Sulfur-annealed Cu₂SnS₃ (CTS) thin films for solar cell applications. *J. Mater. Sci. Mater. Electron.* (2024). <https://doi.org/10.1007/s10854-024-12988-5>
28. Q. Zhang, H. Deng, J. Yu, B. Xu, J. Tao, P. Yang et al., Effect of sulfurization temperature of solution-processed Cu₂SnS₃ absorber for low cost photovoltaic cells. *Mater. Lett.* **228**, 447–449 (2018). <https://doi.org/10.1016/j.matlet.2018.06.083>
29. G. Yang, X. Li, X. Ji, X. Xu, A. Wang, J. Huang et al., Phase composition of the earth-abundant Cu₂SnS₃ thin films with different annealing temperature and its effects on the performance of the related solar cells. *Sol. Energy* **208**, 206–211 (2020). <https://doi.org/10.1016/j.solener.2020.07.090>
30. X. Liu, X. Li, X. Li, Q. Li, D. Zhang, N. Yu et al., Fabrication of Cu₂SnS₃ thin film solar cells via a sol-gel technique in air. *Physica B* (2022). <https://doi.org/10.1016/j.physb.2021.413613>
31. Y. Zhao, X. Han, B. Xu, C. Dong, J. Li, X. Yan, Effect of sulfurization process on the properties of solution-processed Cu₂SnS₃ thin film solar cells. *J. Mater. Sci. Mater. Electron.* **30**, 17947–17955 (2019). <https://doi.org/10.1007/s10854-019-02148-5>
32. Y. Igarashi, T. Tomono, K. Tanaka, K. Moriya, Preparation of Cu₂SnS₃ thin film by sol-gel dip coating. *Jpn. J. Appl. Phys.* (2022). <https://doi.org/10.35848/1347-4065/ac2e7b>
33. S. Erkan, Y. Atasoy, M.A. Olğar, R. Zan, Niğde Ömer Halisdemir Üniversitesi **14**(1), 36–043 (2025). <https://doi.org/10.28948/ngmuh.1549548>
34. J.M. Correa, R.A. Becerra, A.A. Ramírez, G. Gordillo, Fabrication of solar cells based on Cu₂ZnSnS₄ prepared from Cu₂SnS₃ synthesized using a novel chemical procedure. *EPJ Photovoltaics* (2016). <https://doi.org/10.1051/epjpv/2016005>

35. H. Shang, Y. Zhang, Y. Li, Y. Qi, S. Guo, D. Zhao, Effects of adding over-stoichiometrical Ti and substituting Fe with Mn partly on structure and hydrogen storage performances of TiFe alloy. *Renew. Energy* **135**, 1481–1498 (2019). <https://doi.org/10.1016/j.renene.2018.09.072>
36. P. Paufler, C. S. Barrett, T. B. Massalski. *Structure of Metals*. 3rd revised edition. Pergamon Press Oxford, New York, Toronto, Sydney, Paris Frankfurt/M 1980 654 Seiten, 113 Abbildungen, 19 Tabellen und über 1400 Literaturhinweise. Preis US \$ 20.–. *Kristall Tech* **16**, 982–982 (1981). <https://doi.org/10.1002/crat.19810160904>
37. D.M. Berg, R. Djemour, L. Gütay, S. Siebentritt, P.J. Dale, X. Fontane et al., Raman analysis of monoclinic Cu₂SnS₃ thin films. *Appl. Phys. Lett.* (2012). <https://doi.org/10.1063/1.4712623>
38. A.C. Lokhande, P.T. Babar, V.C. Karade, M.G. Gang, V.C. Lokhande, C.D. Lokhande et al., The versatility of copper tin sulfide. *J. Mater. Chem. A Mater.* **7**, 17118–17182 (2019). <https://doi.org/10.1039/C9TA00867E>
39. T.R. Rana, S.Y. Kim, J.H. Kim, Existence of multiple phases and defect states of SnS absorber and its detrimental effect on efficiency of SnS solar cell. *Curr. Appl. Phys.* **18**, 663–666 (2018). <https://doi.org/10.1016/j.cap.2018.03.024>
40. D. Alagarasan, S.S. Hegde, S. Varadharajaperumal, K.D. Arun Kumar, R. Naik, S.P. Panjalingam et al., Effect of annealing temperature on SnS thin films for photodetector applications. *J. Mater. Sci. Mater. Electron.* **33**, 4794–4805 (2022). <https://doi.org/10.1007/s10854-021-07668-7>
41. Y. Atasoy, Effect of annealing temperature on the microstructural and optical properties of newly developed (Ag,Cu)₂Zn(Sn,Ge)Se₄ thin films. *Appl. Phys. A* (2022). <https://doi.org/10.1007/s00339-022-06188-3>
42. L.L. Baranowski, P. Zawadzki, S. Christensen, D. Nordlund, S. Lany, A.C. Tamboli et al., Control of doping in Cu₂SnS₃ through defects and alloying. *Chem. Mater.* **26**, 4951–4959 (2014). <https://doi.org/10.1021/cm501339v>
43. P.G. Suryawanshi, B.M. Babar, A.A. Mohite, U. Pawar, A.G. Bhosale, H.D. Shelke, A simple chemical approach for the deposition of Cu₂SnS₃ (cts) thin films. *Mater. Today Proc.* **43**, 2682–2688 (2020). <https://doi.org/10.1016/j.matpr.2020.05.211>
44. J. Henry, K. Mohanraj, S. Kannan, S. Barathan, G. Sivakumar, Structural and optical properties of SnS nanoparticles and electron-beam-evaporated SnS thin films. *J. Exp. Nanosci.* **10**, 78–85 (2015). <https://doi.org/10.1080/17458080.2013.788226>
45. F. Rehman, S. Yılmaz, İ. Polat, E. Bacaksız, R. Zan, M.A. Olgar, Unveiling improvements in structural, optical, and photodetection characteristics of CZTS thin films through Ag-doping. *Opt. Mater.* (2025). <https://doi.org/10.1016/j.optmat.2025.117006>
46. J.I. Pankove, D.A. Kiewit, Optical processes in semiconductors. *J. Electrochem. Soc.* **119**, 156C (1972). <https://doi.org/10.1149/1.2404256>
47. O.G. Abdullah, S.B. Aziz, K.M. Omer, Y.M. Salih, Reducing the optical band gap of polyvinyl alcohol (PVA) based nanocomposite. *J. Mater. Sci. Mater. Electron.* **26**, 5303–5309 (2015). <https://doi.org/10.1007/s10854-015-3067-3>
48. L. Xi, Y.B. Zhang, X.Y. Shi, J. Yang, X. Shi, L.D. Chen et al., Chemical bonding, conductive network, and thermoelectric performance of the ternary semiconductors Cu₂SnX₃ (X= Se, S) from first principles. *Phys. Rev. B* **86**, 155201 (2012). <https://doi.org/10.1103/PhysRevB.86.155201>
49. J. Wang, J. Chen, L. Zhang, Z. Zhang, Y. Han, X. Hu et al., Forming mechanism and growth of Kirkendall voids of Sn/Cu joints for electronic packaging: a recent review. *J Adv Join Process* (2022). <https://doi.org/10.1016/j.jajp.2022.100125>
50. V. Pavan Kumar, P. Lemoine, V. Carnevali, G. Guélou, O. Lebedev, P. Boullay et al., Ordered sphalerite derivative Cu₅Sn₂S₇: a degenerate semiconductor with high carrier mobility in the Cu–Sn–S diagram ordered sphalerite derivative Cu₅Sn₂S₇: a degenerate semiconductor with high carrier mobility in the Cu. *J. Mater. Chem. A Mater.* n.d. **9**, 10812–10826 (2021)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.